OOPS ASSIGNMENT

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Batch - 2O23

Q1 -

```
#include <iostream>

#include <math.h>

#include <string>

using namespace std;


struct students{

string name;

long roll;

string degree;

char hostel;

float cgpa;

void addDetails(){

cout<<"Enter Name: ";

cin>>name;

cout<<"Enter Roll.No: ";

cin>>roll;

cout<<"Enter Degree: ";
```

```
cin>>degree;
cout<<"Enter hostel: ";
cin>>hostel;
cout<<"Enter cgpa: ";
cin>>cgpa;
}
void updateDetails(){
cout<<"Enter new Name: ";
cin>>name;
cout<<"Enter new Roll.No: ";
cin>>roll;
cout<<"Enter new Degree: ";
cin>>degree;
cout<<"Enter new hostel: ";
cin>>hostel;
cout<<"Enter new cgpa: ";
cin>>cgpa;
}
void updateCGPA(){
cout<<"Enter new CGPA: ";
cin>>cgpa;
}
void updateHostel(){
cout<<"Enter new Hostel: ";
```

```
cin>>hostel;

}

void displayDetails(){

cout<<name<<endl;

cout<<roll<<endl;

cout<<degree<<endl;

cout<<hostel<<endl;

cout<<cgpa<<endl;

}

};



int main(){

students s;

s.addDetails();

s.displayDetails();

s.updateHostel();

s.displayDetails();

s.updateCGPA();

s.displayDetails();

return 0;

}
```

```
(base) vihaan@Vihaans-Macbook Desktop % cd "/Users/vihaan/Desktop/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/"trash
Enter Name: Vihaan
Enter Roll.No: 1024150189
Enter Degree: ENC
Enter hostel: D
Enter cgpa: 9.33
Vihaan
1024150189
ENC
D
9.33
Enter new Hostel: M
Vihaan
1024150189
ENC
M
9.33
Enter new CGPA: 10
Vihaan
1024150189
ENC
M
10
(base) vihaan@Vihaans-Macbook Desktop %
```

Q2 -

Public members can be accessed directly from outside the class using an object, while private members cannot be accessed directly and can only be used inside the class through member functions. This helps in data hiding and protects the internal data of the class from accidental modification.

```
#include <iostream>

#include <string>

using namespace std;



class Student {

private:

    string name;

    int roll;

    string degree;

    string hostel;
```

```
    float cgpa;


    void updateCGPA() {
        cin >> cgpa;
    }


    void updateHostel() {
        cin >> hostel;
    }


public:
    void addDetails() {
        cin >> name >> roll >> degree >> hostel >> cgpa;
    }


    void updateDetails() {
        updateCGPA();
        updateHostel();
    }


    void displayDetails() {
        cout << name << endl;
        cout << roll << endl;
        cout << degree << endl;
```

```
        cout << hostel << endl;

        cout << cgpa << endl;

    }

};



int main() {

    Student s;

    s.addDetails();

    s.updateDetails();

    s.displayDetails();

    return 0;

}
```

```
(base) vihaan@Vihaans-Macbook Desktop % cd "/Users/vihaan/Desktop/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/"trash
Vihaan
1024150189
ENC
D
9.33
10
M
Vihaan
1024150189
ENC
M
10
(base) vihaan@Vihaans-Macbook Desktop %
```

Q3 -

```
#include <iostream>

using namespace std;



class Demo {

private:
```

```
void pvtFunction() {

cout << "My name is Vihaan!" << endl;

}



public:

void publicFunction() {

pvtFunction();

}

};



int main() {

Demo obj;

obj.publicFunction();

return 0;

}
```

```
(base) vihaan@Vihaans-Macbook Desktop % cd "/Users/vihaan/Desktop/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/"trash
My name is Vihaan!
(base) vihaan@Vihaans-Macbook Desktop %
```

Q4 -

```
#include <iostream>

using namespace std;
```

```
class Rectangle {

int width;

int height;


public:

void getdata() {

cin >> width >> height;

}


void calculatearea() {

cout << width * height;

}

};


int main() {

Rectangle r;

r.getdata();

r.calculatearea();

return 0;

}


#include <iostream>

using namespace std;
```

```
class Rectangle {

int width;

int height;


public:

void getdata() {

cin >> width >> height;

}


void calculatearea() {

cout << width * height;

}

};


int main() {

Rectangle r;

r.getdata();

r.calculatearea();

return 0;

}
```

```
(base) vihaan@Vihaans-Macbook Desktop % cd "/Users/vihaan/Desktop/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/"trash
12
13
156%
(base) vihaan@Vihaans-Macbook Desktop %
```

Q5 -

```
#include <iostream>
using namespace std;

class Complex {
float real;
float imag;

public:
void setComplex(float r, float i) {
real = r;
imag = i;
}

void displayComplex() {
cout << real << " + " << imag << "i" << endl;
}

Complex addComplex(Complex c) {
Complex temp;
temp.real = real + c.real;
temp.imag = imag + c.imag;
return temp;
```

```
}

};



int main() {

Complex c1, c2, c3;



c1.setComplex(2.5, 3.5);

c2.setComplex(1.5, 2.5);



c3 = c1.addComplex(c2);

c3.displayComplex();



return 0;

}
```

```
(base) vihaan@Vihaans-Macbook Desktop % cd "/Users/vihaan/Desktop/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/"trash
4 + 6i
(base) vihaan@Vihaans-Macbook Desktop %
```

Q6 -

```
#include <iostream>

using namespace std;



int x = 100;
```

```
class Demo {
public:
    static int count;
    void show();
};

int Demo::count = 10;

void Demo::show() {
    int x = 50;
    cout << "Local x: " << x << endl;
    cout << "Global x: " << ::x << endl;
    cout << "Static count: " << Demo::count << endl;
    std::cout << "Using scope resolution with cout" << std::endl;
}

int main() {
    Demo d;
    d.show();
    return 0;
}
```

```
(base) vihaan@Vihaans-Macbook Desktop % cd "/Users/vihaan/Desktop/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/"trash
Local x: 50
Global x: 100
Static count: 10
Using scope resolution with cout
(base) vihaan@Vihaans-Macbook Desktop %
```

Q7 -

```
#include <iostream>
using namespace std;


namespace SectionA {
    int value = 10;
    void display() {
        cout << "SectionA value: " << value << endl;
    }
}


namespace SectionB {
    int value = 20;
    void display() {
        cout << "SectionB value: " << value << endl;
    }
}


int main() {
    SectionA::display();
```

```
    SectionB::display();

    return 0;

}
```

```
(base) vihaan@Vihaans-Macbook Desktop % cd "/Users/vihaan/Desktop/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/"trash
SectionA value: 10
SectionB value: 20
(base) vihaan@Vihaans-Macbook Desktop %
```